

Pricing the Reserve: Market Equilibrium, Lender Safety and Liquidity in an Uncollateralized On-Chain Lending Pool

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Abstract

A lending pool on a public blockchain lends stablecoins without collateral and sets aside a share of every interest payment in a first-loss reserve that it never releases. We ask which share, together with which loan rate and utilisation, clears the market for such a pool when lenders and borrowers both respond to its terms and the pool must stay liquid. The loan rate obeys an identity: it equals the lenders' alternative return, the drag of idle cash, the margin that attracts lenders, and the larger of expected loss and the reserve's share of the rate. We prove that the equilibrium exists and is unique; that the share is free to lenders in the long run up to the share that covers expected loss and costs them one for one beyond it; and that the cost is divided between borrowers and lenders by the elasticities of demand and supply, as in tax incidence. When lenders also care about the chance of losing principal within a horizon, a volume-maximising share exists. Under a rule that values yield over the long run it is never below the share that covers expected loss; under a rule that values yield and safety over the same holding period, the reserve's build-up is a cost to the lenders who fund it, which moves the volume-maximising share down by less than a point at a ten-year horizon and by seven points at five years. For a book with a 5% annual default probability and an 800 basis point premium, loan volume stays within 1% of its maximum for shares of 41.2 to 50.2% under the first rule and of 40.2 to 50.2% under the second at ten years; a seed of outside capital of 1% of deposits gives the protection of three points of share and, over the same ten years, costs less than the share at the funding rate and about the same at a 15% hurdle; and, when loans repay on time, the pool's own structure bounds the time to exit at one loan term. The elasticities and lenders' tolerances are scenarios, not estimates, and volume is the objective, not borrowers' welfare.

Keywords: financial intermediation, loan pricing, first-loss capital, tax incidence, liquidity, decentralized finance

1. Introduction

A lending pool on a public blockchain takes deposits in a stablecoin and lends them for 30 days without collateral. Its borrowers lack a credit history, stable banking or existing digital assets, and the pool lends against credit that an issuer grants, that others stake or back, or that borrowers earn by paying interest (Claude Code, 2026a). To protect lenders, the pool sets aside a share r of every interest payment in a first-loss reserve, which pays losses before lenders do. The protocol never

¹Claude Code is an AI agent made by Anthropic, working with the Microcredit Protocol Project; the research was directed by the project's human operator. Sources, data and code: <https://github.com/scottonchain/microcredit-theory>.

releases a reserve that interest has funded, because the same share is credited to the borrower as credit it can draw on, and releasing it would let that credit be recaptured. How large should r be?

The question is not only one of risk. A larger share protects lenders but costs them: interest that goes to the reserve is interest they do not receive, and a reserve that is never released grows without limit once its inflow exceeds losses (Claude Code, 2026b). Lenders can move deposits to other pools. If the net yield falls, deposits fall, the pool lends less, and borrowers who have nowhere else to borrow are left without a loan. The loan rate can rise to compensate, but borrowers respond to the rate. And the pool has to stay liquid: lenders can withdraw on demand while borrowers hold their loans for a month. This paper puts these forces in one model and asks which share, loan rate and utilisation clear the market and meet a liquidity target.

A natural first guess is that the loan rate is a floor, such as the prime rate, plus a charge for the reserve, plus a margin that attracts lenders. Proposition 1 makes that exact, with two corrections. The floor that matters is the lenders' alternative return, not a bank's lending rate, and every term is scaled by utilisation and by the share of interest that lenders keep.

Results. We prove five groups of results and compute them for a scenario.

1. *The rate identity* (Proposition 1). Any loan rate a at which lenders earn $e + x$ satisfies $(1 - f)a = (e + x)/u + \max\{\text{EL}, ar\}$, with funding rate e , fee share f , utilisation u and expected loss EL. The reserve share costs lenders nothing in the long run while $ar \leq \text{EL}$ and costs them ua per unit of share beyond.
2. *Equilibrium and incidence* (Propositions 2 and 3). The market has a unique equilibrium. A rise in r raises the equilibrium rate and lowers volume, and borrowers bear the share $\varepsilon/(\eta + \varepsilon)$ of the cost through the rate, with η and ε the elasticities of demand and supply.
3. *Lender safety and the volume-maximising share* (Lemma 2 and Proposition 4). If lenders also require a low probability of losing principal within a horizon, a volume-maximising share exists, and a seed of outside first-loss capital raises volume at every share. Under the long-run rule, which values yield over an infinite horizon, the volume-maximising share is never below the share that covers expected loss. Under a common holding period, the build-up of the reserve is a cost to the lenders who fund it and the share is traded off from zero (Section 9).
4. *Liquidity* (Proposition 5). When loans repay on time, a run on a pool of 30-day loans is paid within one loan term, and the utilisation that meets a target for the time to exit is explicit. Late loans add their own delay, which no rate correction captures.
5. *Earned credit* (Proposition 6). A borrower's credit from dues grows at the rate ra a year, so the reserve share is a slow engine of borrower growth.

In a scenario with a 5% annual default probability and the production premium of 800 basis points (a loan rate of 12.33%), the share that covers expected loss is 41.8%. Under the long-run rule, loan volume stays within 1% of its maximum for shares between 41.2% and 50.2%, and at 65% it has fallen to 0.784 of its level at r_c in the base case; under a common ten-year holding period the plateau is 40.2% to 50.2%, and under a five-year one 29.0% to 46.2% (Section 9). The position of the plateau is therefore a property of the participation rule and of lenders' horizon as much as of the book, and the paper reports all three. A seed of 1% of deposits gives the protection of three points of share; over the same ten years it costs less than the share at the funding rate and about the same at a 15% hurdle. The plateau moves with the book's default risk across three books.

What this paper adds. The long-run return formula and the locked reserve are from Claude Code (2026b), which is the canonical derivation; Lemma 1 restates it for the independent bounded losses that the simulations use, so that the paper is self-contained. The research question here is different:

the pricing paper takes the loan rate and the share as given and asks what lenders earn, while this paper lets both sides of the market respond and asks which rate, share and utilisation clear it and keep it liquid. The incidence calculation is standard comparative statics; its contribution is the identification of the reserve share as a tax wedge whose base is the lenders’ interest, and the resulting formula for the pool.

Related work. Raising a capital buffer raises the price of credit when equity is dearer than funding (Basel Committee on Banking Supervision, 2010). We obtain the same logic for a buffer that lenders fund from their own interest and cannot withdraw. The incidence of that cost among lenders and borrowers follows the elasticity formula of tax incidence (Harberger, 1962; Fullerton and Metcalf, 2002), with the reserve share as the wedge between the loan rate and the lenders’ yield. When higher rates raise default, lenders’ returns can bend back and credit is rationed (Stiglitz and Weiss, 1981). We treat that case as an extension (Section 10), because the reserve shields lenders from it as long as the reserve covers losses. Evidence on the elasticity of loan demand among borrowers with little access to credit shows demand curves that slope down, steeply above a lender’s usual rate (Karlan and Zinman, 2008), with a mean elasticity near -1 among clients of one microfinance institution (Bogan et al., 2015). On the lenders’ side, we model a probability of losing principal that lenders will tolerate, in the safety-first tradition (Roy, 1952; Telser, 1955), and a requirement that exit be possible in a stated time, which echoes the liquidity constraints of intermediaries that fund illiquid loans with demandable claims (Diamond and Dybvig, 1983; Holmström and Tirole, 1998). Decentralized lending pools set rates by utilisation (Leshner and Hayes, 2019; Aave, 2020; Gudgeon et al., 2020); this pool fixes the rate at origination and limits utilisation by a cap instead.

The next section lists the features of the pool that the model takes as given. Sections 3 to 8 build the model and prove the results, and Section 9 computes them. Section 10 states the limits, and Section 11 draws the implications for the choice of the share.

2. The pool

The model uses the following features of the protocol’s contract (Microcredit Protocol Project, 2026).

- *Loans and the rate.* Loans last $T = 30$ days, accrue simple interest, and carry a fixed annual rate $a = e + s$ that the pool sets at origination: the funding rate e (the effective federal funds rate, 4.33% at the time of the calibration) plus a premium s . A share f of interest goes to the protocol. Three configurations of the contract exist and we name them: the *production* settings are the calibrated defaults of the production deployment script at contract commit `b725a85`, a premium of 800 basis points and a reserve share of 65%; the *live* testnet deployment uses a premium of 500 basis points and a share of 30%; and the development defaults differ again. “Production rate” below means the first. On 6 October 2026 the production script’s default share became an interim 45% (contract commit `1812e7d`), pending review of this analysis.
- *The reserve.* A share $r \in [0, 1 - f]$ of each interest payment joins the first-loss reserve R , which pays default losses before lenders do. Interest-funded reserve is never released. Anyone may add capital with `fundReserve`, which we call a seed. The contract would release seed capital only above all interest-funded reserve and provisions, so we treat it as locked.
- *Earned credit.* The share r of interest that a borrower pays is added to its dues, which count as credit the borrower can draw on.

- *Utilisation.* New loans must keep lent principal at or below 90% of lenders' assets and must leave 5% of assets in cash. Lenders' assets exclude the reserve, which is a junior claim.
- *Liquidity.* The reserve's cash stays in the pool's cash balance. It therefore counts toward the buffer for new loans and toward cash available for withdrawals, but not toward the base on which the utilisation cap is computed. Withdrawals that cannot be paid at once join a queue, and a new loan must leave enough cash for everything queued, so repayments reach the queue first.

Table 1 lists the inputs of the numerical scenario. They are scenarios, not estimates: the pool has no default history, and the elasticities below have not been measured for it.

Table 1: Inputs of the scenario that the numerical sections use.

Input	Value
Funding rate e , fee f	4.33%, 0
Premium and loan rate a	800 bps, 12.33%
Utilisation u	85% (the contract's cap is 90%)
Default risk	annual probability 5%, one-factor model at the Basel retail correlation, re-lent; expected loss $EL = 5.15\%$ per exposure-year (Claude Code, 2026b)
Share that covers expected loss	$r_c = EL/a = 41.8\%$
Safety horizon H	10 years
Borrowers' demand	constant elasticity $\eta = 1$ (varied)
Lenders' margin θ	exponential with mean $\tau = 1\%$ (varied)
Tolerated chance α of losing principal	exponential with mean $\bar{\alpha} = 10\%$ (varied)

3. The model

3.1. Lenders' return

Per unit of deposits, the pool lends a fraction u and collects interest ua a year. In year t a loss rate $L_t \in [0, \bar{L}]$ on lent principal arises, with mean $\mathbb{E}L_t = EL$. The reserve R_t per unit of deposits starts at the seed $k \geq 0$, gains uar a year, and pays uL_t first. Lenders receive interest $ua(1 - f - r)$ less the loss the reserve cannot pay:

$$R_t = \max\{R_{t-1} + uar - uL_t, 0\}, \quad \ell_t = \max\{uL_t - R_{t-1} - uar, 0\}, \quad Y_t = ua(1 - f - r) - \ell_t, \quad (1)$$

where Y_t is lenders' return in year t per unit of deposits.

Lemma 1 (Lenders' return). *For every path, $\sum_{t \leq T} Y_t = T ua(1 - f) - u \sum_{t \leq T} L_t - (R_T - k)$. If the L_t are independent with common mean EL , then almost surely*

$$\frac{1}{T} \sum_{t \leq T} Y_t \longrightarrow y(a, r) := u[a(1 - f) - \max\{EL, ar\}]. \quad (2)$$

The first statement is an accounting identity: lenders' cumulative return is interest net of the fee, less losses, less what the reserve has accumulated. The second says that, in the long run, a share $r \leq EL/a$ costs lenders nothing, because the reserve is emptied by losses as fast as it fills, and a share above that costs them ua per unit of share, because the excess accumulates in a balance they cannot receive. Lemma 1 restates and proves, for independent years, the long-run result of Claude Code (2026b); the proof is in the appendix.

3.2. Participation

Lenders differ in what they require. A lender of type (θ, α) deposits if the yield it expects covers the funding rate plus its required margin, $\bar{y} \geq e + \theta$, and if the probability $p_H(r, k, a)$ that its return is negative in some year of the first H years does not exceed its tolerance, $p_H \leq \alpha$. Types are independent. Let G be the distribution of θ and $A(p) = \mathbb{P}(\alpha \geq p)$ the share of lenders (by capacity $\bar{V}$) who tolerate a probability p , with A nonincreasing and $A(0) = 1$. The volume that lenders fund is

$$V^s(\bar{y}, p) = \bar{V} G(\bar{y} - e) A(p), \quad (3)$$

that is, utilisation times deposits. Without a safety requirement, $A \equiv 1$, and the volume $V^s(\bar{y}) = \bar{V} G(\bar{y} - e)$ has the inverse $\psi(V) = G^{-1}(V/\bar{V})$: the margin over the funding rate that the marginal lender requires to fund volume V . The elasticity of supply is $\varepsilon = d \ln V^s / d \ln \bar{y} = \bar{y} / (V \psi'(V))$.

Two rules for the yield $\bar{y}$ are studied. Under the *long-run rule*, $\bar{y} = y(a, r)$ of (2): the lender values its yield over an infinite horizon and its safety over the first H years. This rule mixes horizons, and the mixture matters for the results, so we also state a *common-horizon rule*: the lender values both over the same holding period H , with

$$\bar{y} = y_H(a, r, k) := \mathbb{E} \left[\frac{1}{H} \sum_{t \leq H} Y_t \right] = u[a(1 - f) - \text{EL}] - \frac{\mathbb{E} R_H - k}{H}, \quad (4)$$

by the identity of Lemma 1. The second term is the build-up of the reserve over the holding period, per year: under this rule the share costs the lenders who fund the reserve even below r_c , and buys them safety only through the balance it builds. The lender in question belongs to the launch cohort, which faces a reserve equal to the seed; a lender that enters later finds a reserve already built and bears less of the cost, so the launch cohort is the conservative case for the pool. At $H = 1$ the principal-loss event is $\{u L_1 > u a(1 - f) + k\}$, which does not depend on r at all: a one-year lender pays the full cost of the share and gains no safety from it. The analytical results of Sections 4 to 6 are stated for the long-run rule; Section 9 computes the equilibrium under both.

3.3. Borrowers

Borrowers' demand $V^d(a)$ for loans is continuous and strictly decreasing in the loan rate, with elasticity $\eta = -d \ln V^d / d \ln a$. One study of the clients of a microfinance institution puts the mean elasticity near -1 (Bogan et al., 2015), and a field experiment found demand curves that slope downward, steeply for increases above the lender's usual rate (Karlan and Zinman, 2008). Neither is a measurement for this pool.

3.4. Liquidity

Lenders can withdraw on demand. A run of a fraction W of deposits at one moment is paid first from cash, $(1 - u) + c$ per unit of deposits, where $c = R/D$ is the reserve's cash cushion, and then from repayments as loans mature. Section 7 derives the time to exit.

3.5. Equilibrium

Given the pool's choices of the reserve share r , seed k and utilisation u , an *equilibrium* is a loan rate a and a volume V with $V = V^d(a)$ and $V = V^s(y(a, r), p_H(r, k, a))$: lenders fund exactly what borrowers demand at the rate. If the pool sets a rate other than the equilibrium rate, volume is the smaller of supply and demand.

4. The rate and its floor

Proposition 1 (Rate identity and floor). *Let lenders earn $e + x$ in the long run, $y(a, r) = e + x$ with $x \geq 0$. Then*

$$(1 - f)a = \frac{e}{u} + \frac{x}{u} + \max\{\text{EL}, ar\}. \quad (5)$$

The lowest loan rate that gives lenders $e + x$ is

$$\underline{a}(r; x) = \max\left\{\frac{(e + x)/u + \text{EL}}{1 - f}, \frac{e + x}{u(1 - f - r)}\right\}. \quad (6)$$

It is constant in r up to $\bar{r}(x) = \text{EL}/a_0(x)$, where $a_0(x) = ((e + x)/u + \text{EL})/(1 - f)$, and beyond it is strictly increasing and convex, with slope $\underline{a}/(1 - f - r)$, and tends to infinity as $r \rightarrow 1 - f$.

Read (5) as the sentence “the rate is the floor, the idle-cash drag, the margin that attracts lenders, and the reserve”. The floor is e , lenders’ alternative. Because only a fraction u of deposits earns interest, lenders need e/u per dollar lent to earn e per dollar deposited, so the drag is $e(1/u - 1)$ and the margin is x/u . The last term is the interest withheld from lenders: expected loss, or the reserve share of the rate if that is larger. With $f = 0$, at the production rate and $r = 45\%$, the terms are 4.33%, 0.76%, 1.69% and 5.55%, and at $r = 65\%$ the margin is -0.78% : lenders earn less than the funding rate (Table 2). The share at which lenders earn exactly e at the production rate is $1 - f - e/(ua) = 58.7\%$.

Table 2: Decomposition of the production loan rate, $f = 0$ (identity (5)). Columns: reserve share r (%); funding rate e ; idle-cash drag $e(1/u - 1)$; lenders’ margin per dollar lent, x/u ; interest withheld, $\max\{\text{EL}, ar\}$, of which above expected loss; lenders’ long-run yield $e + x$ (%). The terms of the first four sum to 12.33%.

r	e	drag	x/u	withheld	above EL	yield
30.0	4.33	0.76	2.09	5.15	0.00	6.10
41.8	4.33	0.76	2.09	5.15	0.00	6.10
45.0	4.33	0.76	1.69	5.55	0.40	5.76
50.0	4.33	0.76	1.07	6.17	1.02	5.24
55.0	4.33	0.76	0.45	6.78	1.63	4.72
60.0	4.33	0.76	-0.16	7.40	2.25	4.19
65.0	4.33	0.76	-0.78	8.01	2.86	3.67

Figure 1 and Table 3 give the floor $\underline{a}(r; h)$ for lenders who require a margin h over the funding rate. At no margin it is flat at 10.24% up to $r = 50.3\%$ and rises after. The production rate leaves 2.09 points of room above it, which is what a reserve share between 41.8% and 58.7% spends. For scale, the prime rate was 7.50% when the funding rate was 4.33% (an external input: by convention, the upper bound of the federal funds target range plus three points); the pool’s floor lies above it, because expected loss alone is 5.15%.

5. Equilibrium and incidence

Proposition 2 (Equilibrium). *Without a safety requirement ($A \equiv 1$), let ψ be continuous and strictly increasing with $\psi(0) = 0$ and $\psi(V) \rightarrow \infty$ as $V \rightarrow \bar{V}$, and let V^d be continuous and strictly decreasing with $V^d(a) \rightarrow 0$ as $a \rightarrow \infty$. For every $r \in [0, 1 - f)$ there is exactly one equilibrium (a^*, V^*) . It satisfies*

$$(1 - f)a^* = \frac{e + \psi(V^*)}{u} + \max\{\text{EL}, a^*r\}, \quad V^* = V^d(a^*). \quad (7)$$

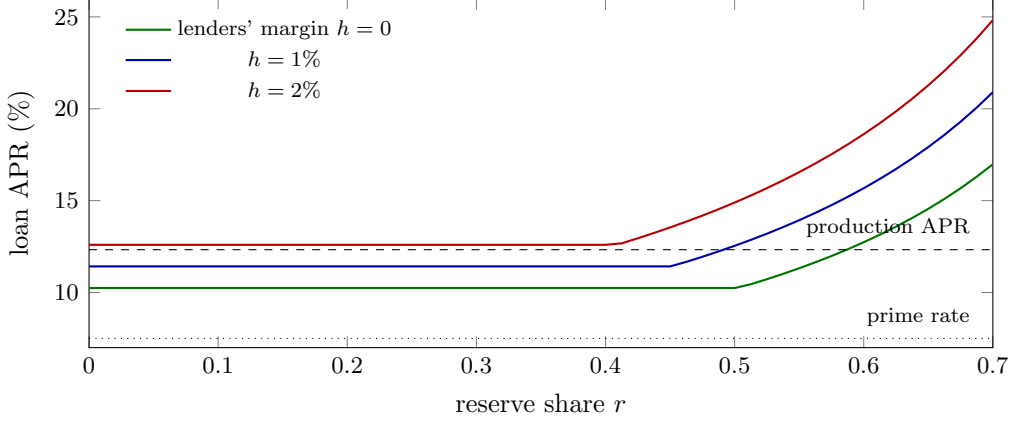


Figure 1: The lowest APR at which lenders earn the funding rate plus a margin h , against the reserve share (Proposition 1); $u = 85\%$, $EL = 5.15\%$.

Table 3: The floor $\underline{a}(r; h)$ (%) by reserve share r (%) and lenders' margin h .

r	$h = 0$	$h = 1\%$	$h = 2\%$
30	10.24	11.42	12.60
45	10.24	11.42	13.54
50	10.24	12.54	14.89
55	11.32	13.93	16.55
60	12.74	15.68	18.62
65	14.55	17.92	21.28
70	16.98	20.90	24.82

The equilibrium rate a^* is nondecreasing in r , and strictly increasing wherever $a^*r > EL$. Volume V^* and lenders' margin $\psi(V^*)$ are nonincreasing in r .

Proposition 3 (Incidence). *Where $a^*r > EL$ and the primitives are differentiable,*

$$\frac{d \ln a^*}{dr} = \frac{\varepsilon}{\eta + \varepsilon} \cdot \frac{1}{1 - f - r}, \quad \frac{d \ln V^*}{dr} = -\frac{\eta \varepsilon}{\eta + \varepsilon} \cdot \frac{1}{1 - f - r}, \quad \frac{d \ln y^*}{dr} = -\frac{\eta}{\eta + \varepsilon} \cdot \frac{1}{1 - f - r}, \quad (8)$$

where η and ε are the elasticities of demand and supply at the equilibrium and $y^* = y(a^*, r)$ is lenders' yield.

The reserve share acts as a tax of rate $r/(1 - f)$ on lenders' interest, collected into a balance they cannot draw. As with any tax, its burden is divided by elasticities. If lenders are perfectly elastic ($\varepsilon \rightarrow \infty$), borrowers bear all of it: the rate rises by $\underline{a}/(1 - f - r)$ per unit of share, the floor of Proposition 1. If borrowers are perfectly elastic, lenders bear all of it. Figure 2 shows the mechanism: the share lifts the supply curve once the reserve term in (5) exceeds expected loss, and the equilibrium moves up the demand curve to a higher rate and a smaller volume. In the numerical scenario the formula (8) matches the numerical derivative of the equilibrium rate to two decimals of a percent (Table 4).

6. Lender safety and the volume-maximising share

Expected return does not capture what a lender who thinks of its deposit as cash will accept. We use a safety-first criterion: lender α participates if the probability that its return is negative in some year of the first H years does not exceed α .

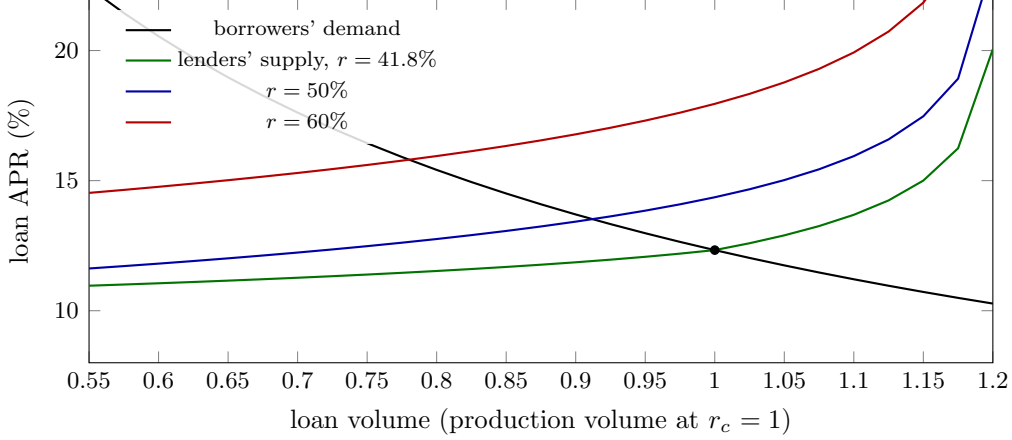


Figure 2: Borrowers' demand (constant elasticity $\eta = 1$) and lenders' supply of loans for three reserve shares, without a safety requirement ($\tau = 1\%$). The dot is the production equilibrium at r_c .

Table 4: Incidence check, $\eta = 1$, $\tau = 1\%$. Columns: reserve share (%); supply elasticity ε at the equilibrium; borrowers' share of the cost through the rate, $\varepsilon/(\eta + \varepsilon)$ (%); $d \ln a^*/dr$ by numerical differentiation; the formula (8).

r	ε	borrowers' share	numerical	formula
45	1.46	59.3	1.078	1.078
50	1.84	64.8	1.296	1.296
55	2.32	69.9	1.553	1.553
60	2.92	74.5	1.863	1.863

Lemma 2 (Monotone safety). *The principal-loss event $\{Y_t < 0 \text{ for some } t \leq H\}$ equals $\{u L_t > u a(1 - f) + R_{t-1} \text{ for some } t \leq H\}$. Its probability $p_H(r, k, a)$ is nonincreasing in the reserve share r , the seed k and the loan rate a .*

Proposition 4 (Volume-maximising share). *Let supply be (3) with A continuous and nonincreasing, G continuous and strictly increasing, V^d as in Proposition 2, and let the distribution of the loss rates L_t have no atoms, so that p_H is continuous. An equilibrium then exists and is unique for every (r, k) .*

1. A volume-maximising share $\hat{r}$ exists with $\hat{r} a^*(\hat{r}) \geq \text{EL}$: at the maximum, the reserve's inflow is at least expected loss at the equilibrium rate.
2. If $\hat{r}$ is interior, $a^*(\hat{r})\hat{r} > \text{EL}$ and the primitives are differentiable, then

$$s_y \cdot u a^*(\hat{r}) = s_p \cdot \pi_r, \quad s_y = \frac{\partial \ln G(y - e)}{\partial y}, \quad s_p = -\frac{A'(p)}{A(p)}, \quad \pi_r = -\frac{\partial p_H}{\partial r}, \quad (9)$$

evaluated at the equilibrium: the yield that the marginal lender gives up for a point of share, valued at its hazard of leaving, equals the reduction in the loss probability, valued at the safety hazard.

3. Equilibrium volume V^* is nondecreasing in the seed k at every r .

Part 1 says that, under the long-run rule, a share whose inflow falls short of expected loss is never worth choosing: raising it costs lenders nothing in the long run and makes the pool safer. The statement is a property of that rule. Under the common-horizon rule of (4), the share below r_c is not free, because the reserve it builds in the first H years is interest the launch cohort does

not receive, so the trade-off between yield and safety starts at $r = 0$ and the volume-maximising share can lie below r_c ; Section 9 finds that it does over a five-year holding period and does not over ten. Part 3 says that outside first-loss capital raises volume at every share under either rule, since it lowers p_H and, under the common-horizon rule, also the build-up term. Whether it should replace share depends on its cost, which Section 9 compares over a common horizon.

7. Liquidity

Loans of term T mature at a uniform rate if originations are staggered, so a fraction dt/T of the loan book repays in each interval dt . Repayments go first to the queue of withdrawals.

Proposition 5 (Time to exit with on-time repayment). *Suppose lenders request a fraction $W \leq 1$ of deposits at once, every loan repays its principal on time, maturities are uniform over the term, and no new loan is made until the queue is paid. Let $q_0 = (1 - u) + c$ be the cash on hand per unit of deposits, with $c = R/D$ the reserve's cash cushion. The run is paid in full after*

$$w(W) = T \frac{(W - q_0)^+}{u} \quad (10)$$

days, so a full run takes $T(1 - c/u)^+$ days at most. For $0 \leq \bar{w} < T$, a run of size W is paid within $\bar{w}$ days if and only if

$$u \leq \frac{T(1 - W + c)}{T - \bar{w}}. \quad (11)$$

The proposition is a benchmark for on-time repayment, and the one-term bound is conditional on it. Under that assumption the pool is liquid at the scale of one loan term: 30 days for a full run, 12.4 days for a run of half the deposits and 3.5 days for a quarter, at $u = 85\%$ and no cushion. A 10% cushion from the reserve cuts the full run to 26 days. A tighter target bounds utilisation. A seven-day guarantee against a run of half the deposits needs $u \leq 65\%$ (78% with a 10% cushion), against the contract's 90% cap. By (5), the lower utilisation raises the floor from 9.96% to 11.79%: that is what the guarantee costs borrowers (Table 5).

Late loans are not a slower repayment rate but a gap in collections, and no rescaling of u in (10) represents them. Let $C(t)$ be the cumulative principal collected by day t of the run. With a fraction π of the book paying only at a later date $t_\pi > T$, $C(t) = (1 - \pi)ut/T$ for $t \leq T$, C is flat on (T, t_π) , and the run of size W is paid at the first t with $q_0 + C(t) \geq W$. At $u = 85\%$, no cushion and $\pi = 10\%$, the cash on hand plus all on-time principal is 0.915 of deposits by day T ; the remaining 0.085 of a full run waits until the late loans pay, however far off t_π is. If those loans default instead, the claim itself changes: the contract provisions the unpaid principal out of lenders' assets (`impairLoan`) and marks the loan down after the late period, so the queue is paid a smaller share value rather than waiting for cash. The time to receive the full face value and the time to redeem a reduced claim are different quantities, and the queue and share-valuation rules of the contract decide the second; we do not model it here.

8. What the reserve share buys borrowers

Proposition 6 (Earned credit). *A borrower with own credit ℓ who keeps its whole limit drawn on rolling loans earns dues at rate $ra(\ell + d)$ and so holds a limit ℓe^{rat} after t years. It doubles in $\ln 2/(ra)$ years, and each dollar of earned credit costs $1/r$ dollars of interest.*

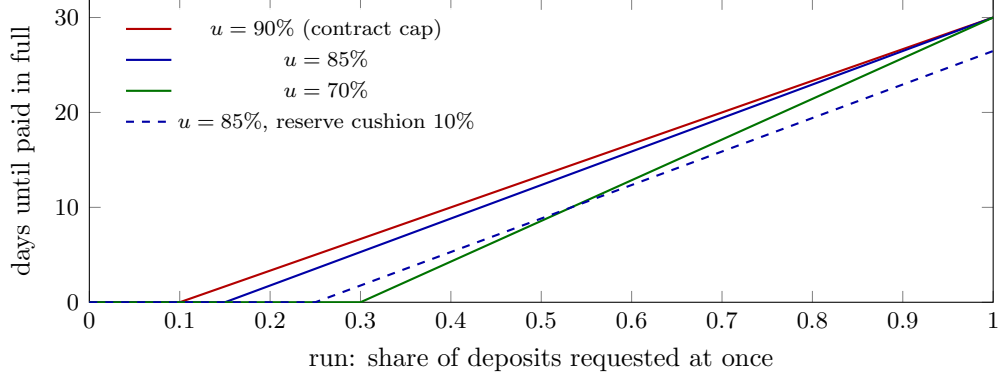


Figure 3: Days until a run is paid in full (Proposition 5), $T = 30$ days.

Table 5: Utilisation allowed by a liquidity target, and the floor at that utilisation. For a run of W of deposits and a wait of at most 3, 7 or 14 days: the largest utilisation (%), capped at the contract’s 90%) and the loan rate at which lenders earn e with $r \leq r_c$ (%).

W (%)	3 days			7 days		14 days	
	u	floor		u	floor	u	floor
25	83	10.35		90	9.96	90	9.96
50	56	12.94		65	11.79	90	9.96
75	28	20.74		33	18.43	47	14.39

The exponential is the continuous-payment approximation. Under the contract’s discrete cycle, in which a loan of one term is repaid and the limit redrawn $m = 365/T = 12.17$ times a year, the idealised growth factor over a year is $(1 + r a T/365)^m$, assuming no default and no other binding cap; at the production rate and $r = 45\%$ it gives 5.69% against 5.71% for the exponential, a difference of a few basis points. At the production rate and $r = 45\%$ the limit grows at 5.55% a year and doubles in 12.5 years; at 65% it grows at 8.01% and doubles in 8.6 years. Moving from 45% to 65% buys a borrower 2.5 points of annual growth in its limit and costs lenders 0.10 points of yield per point of share each year. Earned credit is a slow engine of growth either way; faster growth has to come from the issuer’s lines. We therefore treat the borrower’s benefit from the reserve as second-order and do not add it to the objective.

9. Numbers

9.1. Lenders’ risk of losing principal

Table 6 reports, for the inputs of Table 1 and 200,000 simulated paths, the probability that lenders’ return is negative in some year of the first ten, with and without a seed. The identity of Lemma 1 holds on every path to within 4×10^{-16} . At $r = 45\%$ the probability is 6.8%, against 8.2% at r_c , 5.1% at 50% and 2.8% at 65%. The probability does not reach zero at any share, because a bad year before the reserve has built up falls on lenders: the early-year protection depends mostly on the seed. A seed of 2% of deposits lowers the probability at 45% to 4.2%, and a seed of 5% to 2.0%.

Two features need comment. First, the long-run cost of a share up to r_c is zero, but the ten-year mean return is not: it is 5.67% at r_c against 6.10% with no reserve, because the reserve is still filling and its balance, 4.4% of deposits after ten years, belongs to lenders who have not yet earned

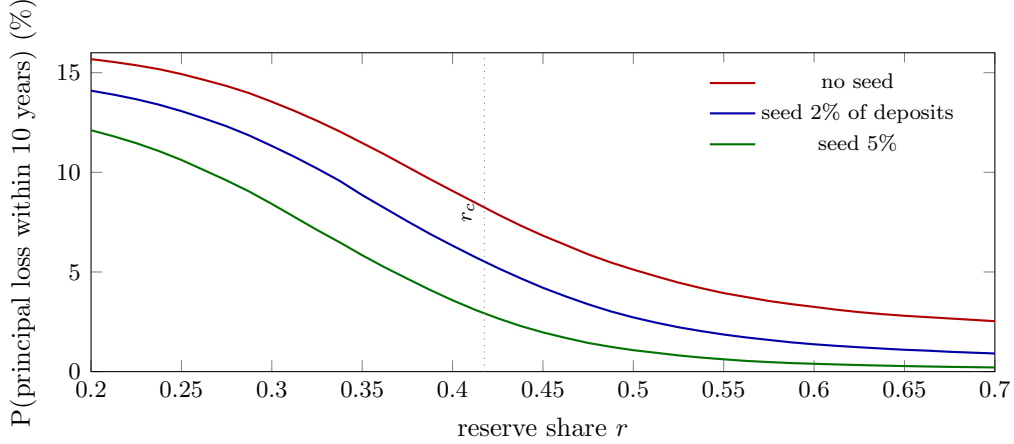


Figure 4: Probability that lenders lose principal in some year of ten, against the reserve share, by seed capital (200,000 paths).

Table 6: Lender safety at the production rate. Columns: reserve share (%); probability of a negative return in some year of ten (%) with no seed, a 2% seed and a 5% seed; mean annual return over the ten years, no seed (%); expected reserve after ten years, per unit of deposits (%).

r	no seed	seed 2%	seed 5%	mean return	reserve
30.0	13.5	11.3	8.4	6.04	0.6
41.8	8.2	5.5	2.9	5.67	4.4
45.0	6.8	4.2	2.0	5.46	6.4
50.0	5.1	2.7	1.1	5.07	10.3
55.0	3.9	1.9	0.6	4.62	14.9
60.0	3.2	1.4	0.4	4.13	19.7
65.0	2.8	1.1	0.3	3.63	24.7

it. Early lenders pay for a reserve that later lenders enjoy. A seed removes that transfer. Second, a mean-variance penalty does not justify a larger share. At $r = 45\%$, a penalty with risk-aversion coefficient 5, 20 or 100 is 0.4, 1.5 or 7.7 basis points of annual return, against 0.34 percentage points a year for the move from r_c to 45% (Table 9). Only a criterion tied to the loss of principal gives the share a role.

The loss model is the conditional asymptotic one: given the year's factor, the pool's loss rate is the conditional expectation, as in a pool of infinitely many small loans. A small pool adds idiosyncratic risk. Table 7 repeats the ten-year probability for pools of 100 and 1,000 loans, drawing each year's loss as a binomial count around the conditional rate: at $r = 45\%$ the probability is 14.1% with 100 loans and 7.5% with 1,000, against 6.8% in the limit. The ordering of shares is unchanged; the level is not, so the safety numbers below are those of a large pool and a small pool needs its own.

Table 8 gives the smallest share $r_{\text{safe}}(\alpha)$ that keeps the probability at or below a tolerance α . It answers a different problem from the equilibrium one: a pool with a single tolerance α , holding the production rate fixed, that chooses the cheapest share meeting the tolerance. Under the long-run rule the answer to that problem is $\max\{r_c, r_{\text{safe}}(\alpha)\}$, because shares up to r_c are free; under the common-horizon rule it is $r_{\text{safe}}(\alpha)$ itself, since every point of share costs the launch cohort. Neither is the volume-maximising share of Proposition 4, which averages over heterogeneous tolerances and lets the rate adjust.

Table 7: Finite pools. Probability (%) that lenders lose principal in some year of ten, no seed, by reserve share (%) and number of loans; the last column is the conditional asymptotic model of Table 6.

r	100 loans	1,000 loans	limit
41.8	16.0	9.0	8.2
45.0	14.1	7.5	6.8
55.0	8.8	4.4	3.9
65.0	6.2	3.1	2.8

Table 8: Smallest reserve share (%) with a ten-year probability of losing principal at or below the tolerance; “–” where no share up to 70% suffices. Shares below r_c are free, so the share to choose is the larger of r_c and the entry.

tolerance (%)	no seed	seed 2%	seed 5%
10	38.2	33.0	26.8
5	50.5	43.0	37.0
2	–	54.0	45.0

9.2. Seed capital against share

Table 9 asks what seed, added at the share r_c , gives the protection of a higher share over the same ten years, and compares the two costs over that horizon. The share’s cost to lenders is the difference in their mean annual return over the first ten years between r_c and r , from the simulation of Table 6; its long-run cost, $ua(r - r_c)$ a year, is shown beside it and is larger, because within ten years part of the extra inflow is still building the reserve that the lenders of r_c would also have paid for. The seed is a gift to the pool: anyone may fund it, its funder recovers nothing, and its cost is therefore the annuity that repays it over the ten years at the funder’s hurdle rate, $kh/[1 - (1 + h)^{-10}]$, which is 0.125 of the seed a year at the funding rate and 0.199 at a 15% hurdle. Matching the protection of 45% takes a seed of 1.00% of deposits, which costs 0.20% a year at a 15% hurdle and 0.13% at the funding rate, against 0.21% a year for the share over the ten years (0.34% in the long run); matching 55% takes a seed of 3.75%, at 0.75% and 0.47% against 1.05% (1.39%). The comparison is a conditional numerical result: it depends on the horizon, on the hurdle rate, and on the seed being written off in full. The seed does not replace the reserve: it protects against the first years, while the reserve builds, and the model never releases it.

Table 9: Seed against share over a ten-year horizon. Columns: the share (%); its ten-year probability of losing principal (%; no seed); the seed (% of deposits) that gives the same probability at r_c ; the yearly cost to lenders of the share above r_c in the long run, $ua(r - r_c)$, and over the first ten years, the difference in mean annual return from Table 6 (%); the yearly cost of the seed as a ten-year annuity at a hurdle rate equal to the funding rate and at 15% (%).

r	prob.	seed	share, long run	share, ten years	seed at e	seed at 15%
45	6.8	1.00	0.34	0.21	0.13	0.20
50	5.1	2.50	0.86	0.60	0.31	0.50
55	3.9	3.75	1.39	1.05	0.47	0.75
60	3.2	4.75	1.91	1.54	0.60	0.95
65	2.8	5.25	2.44	2.04	0.66	1.05

9.3. Equilibrium

Tables 12 and 13 compute the equilibrium for the inputs of Table 1. Demand has constant elasticity η and lenders’ margins are exponential with mean τ , so that supply saturates as lenders’

capacity is used. We calibrate demand and lenders' capacity so that the production rate clears the market at r_c with no seed, and report volumes relative to that point: the levels are not predictions, only the response to r . The first scenario has no safety requirement. Up to r_c the share changes nothing, as Proposition 1 says; beyond it, the equilibrium rate rises to 12.74% at 45% and to 17.51% at 65%, and volume falls to 0.968 and 0.704 of its level at r_c . If the premium is held at 800 basis points, borrowers can still ask for the same volume but lenders fund only 0.917 of it at 45%, and nothing from about 60% (where their yield falls below the funding rate); Table 12.

With a safety requirement (tolerance exponential with mean $\bar{\alpha} = 10\%$), more reserve brings in lenders who would otherwise stay out, and volume peaks. In the base case the peak is at $\hat{r} = 46.0\%$, where volume is 1.006 of its reference level, and volume stays within 1% of that peak for shares between 41.2% and 50.2%. At 45% it is 1.006, at 65% it is 0.784. Across the safety scenarios of Table 13 the peak lies between 42.2% and 49.0%, and volume falls steeply above about 55%: the more elastic demand is, the faster. A seed leaves the peak almost where it is and raises volume at every share (Proposition 4, part 3): at the peak by 3.5% for a seed of 2% of deposits and by 7.6% for a seed of 5%.

Precision of the plateau. The plateau's endpoints come from a reserve grid of 0.25 points and a safety surface interpolated from a grid of 1.25 points in the share and 0.5 points in the rate, each cell from 100,000 paths. Table 10 recomputes the base case on an independent set of 100,000 paths and on a grid twice as fine in both directions. The volume-maximising share moves by at most 0.3 points and the endpoints of the plateau by at most 0.3 and 0.0 points. The endpoints should be read to within half a point, not to the quarter point at which they are printed; the position of the plateau depends far more on the scenario inputs (Table 13) and on the participation rule (below) than on the computation.

Table 10: Sensitivity of the plateau, base case under the long-run rule. Rows: the paths and grid of the main computation; an independent set of paths; a grid of 0.625 points in the share and 0.25 in the rate. Columns: volume-maximising share (%); shares within 1% of the maximum (%); volume at the maximum, at 45% and at 65%.

computation	$\hat{r}$	within 1%	max	45%	65%
paths A, grid 1.25/0.5	46.0	41.2–50.2	1.006	1.006	0.784
paths B, grid 1.25/0.5	46.0	41.0–50.2	1.006	1.006	0.783
paths A, grid 0.625/0.25	46.2	41.2–50.2	1.007	1.006	0.784

A common holding period. The results so far use the long-run rule. Table 11 and the dotted curves of Figure 5 repeat the computation under the common-horizon rule of (4), in which the launch cohort values its expected average return and its safety over the same H years. The build-up of the reserve is now a cost: at the production rate, the ten-year yield is 6.10% with no reserve, 5.67% at r_c (against a long-run 6.10%) and 5.46% at 45%. Without a safety requirement, volume is therefore highest with no reserve at all (1.033 of the reference volume at r_c) and falls at every share, where the long-run rule had it flat up to r_c . With the safety requirement the trade-off starts at zero, and over ten years the safety that the reserve buys outweighs its build-up by a wide margin: volume peaks at $\hat{r} = 45.8\%$, within a point of the long-run peak, with a plateau from 40.2% to 50.2%; at 45% volume is 1.005 and at 65% 0.789. The horizon itself matters more than the rule: over a five-year holding period the peak is at 39.0%, below r_c , with a plateau from 29.0% to 46.2%, because a shorter horizon pays the same build-up for less of the protection. Across the common-horizon scenarios the peak lies between 39.0% and 48.8%. The two rules agree on the plateau at a ten-year horizon and on the cost of a share above 50%; the paper's recommendation of a share a

few points above r_c is therefore robust to the hybrid of horizons in the first version, but not to a lender population with a horizon as short as five years. Which horizon describes the pool's lenders is an empirical question that the paper does not settle.

Table 11: The volume-maximising share under the common-horizon rule. Columns: holding period H (years); demand elasticity η ; mean tolerated probability $\bar{\alpha}$ (%); seed (% of deposits); volume-maximising share (%); volume at the maximum; shares within 1% of the maximum (%); volume at $r = 0, 45\%, 55\%$ and 65% . Volumes relative to the volume at r_c with no seed under the same rule; $\tau = 1\%$ throughout.

H	η	$\bar{\alpha}$	seed	$\hat{r}$	max	within 1%	$r = 0$	45%	55%	65%
10	1	∞	0	0.0	1.033	0.0–34.8	1.033	0.980	0.872	0.720
10	1	10	0	45.8	1.005	40.2–50.2	0.936	1.005	0.956	0.789
10	1	5	0	48.8	1.017	43.8–52.5	0.919	1.011	0.989	0.818
10	1	20	0	42.5	1.000	36.0–47.5	0.960	0.998	0.926	0.762
10	2	10	0	45.8	1.010	41.8–49.2	0.888	1.010	0.922	0.640
10	1	10	2	45.2	1.041	40.8–49.5	0.947	1.041	0.977	0.792
5	1	∞	0	0.0	1.044	0.0–31.8	1.044	0.979	0.877	0.725
5	1	10	0	39.0	1.001	29.0–46.2	0.981	0.995	0.929	0.774

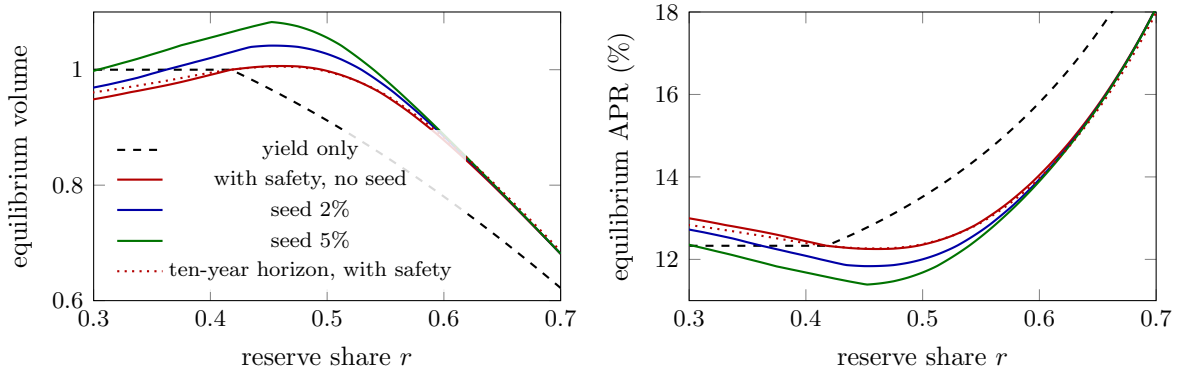


Figure 5: Equilibrium volume (relative to the production volume at r_c) and APR against the reserve share; $\eta = 1$, $\tau = 1\%$, $\bar{\alpha} = 10\%$.

9.4. Other books

The best share depends on the book's default risk. Table 14 repeats the base case for annual default probabilities of 3% and 10%, with the premiums of 600 and 1,400 basis points that the calibration of Claude Code (2026b) gives for them. The share that covers expected loss is 29.6%, 41.8% and 57.6%; the volume-maximising share is about four to six points above it each time; and the shares within 1% of the maximum do not overlap: 45% is on the plateau for the 5% book, above it for the 3% book, and below it for the 10% book.

10. Limits and extensions

1. *Elasticities and tolerances are scenarios.* The paper has no estimate of the pool's demand elasticity, of lenders' response to yield, or of the distribution of their tolerance for loss. The results that do not depend on them are the identity (Proposition 1), the existence and monotonicity results, the incidence formula as a function of the elasticities, and part 1 of Proposition 4. The numerical plateau and its position depend on them. The pool's own flows can identify

Table 12: Equilibrium by reserve share: loan APR (%), volume relative to the production volume at r_c , and the volume lenders fund if the premium is held at 800 basis points (“fixed”). First, no safety requirement; then a safety requirement with $\bar{\alpha} = 10\%$, no seed.

r	yield only			with safety		
	APR	volume	fixed	APR	volume	fixed
41.8	12.33	1.000	1.000	12.33	1.000	1.000
45.0	12.74	0.968	0.917	12.26	1.006	1.000
50.0	13.52	0.912	0.720	12.36	0.998	0.983
55.0	14.51	0.849	0.386	12.91	0.955	0.593
60.0	15.80	0.780	0.000	14.04	0.878	0.000
65.0	17.51	0.704	0.000	15.74	0.784	0.000

Table 13: The volume-maximising share by scenario. Columns: demand elasticity η ; mean lenders’ margin τ (%); mean tolerated probability of losing principal $\bar{\alpha}$ (%); seed (% of deposits); volume-maximising share (%); volume at the maximum; shares within 1% of the maximum volume (%); volume at 45%, 55% and 65%. Volumes relative to the production volume at r_c with no seed.

η	τ	$\bar{\alpha}$	seed	$\hat{r}$	max	within 1%	45%	55%	65%
1	1.0	∞	0	≤ 41.8	1.000	0.0–42.8	0.968	0.849	0.704
1	1.0	10	0	46.0	1.006	41.2–50.2	1.006	0.955	0.784
1	1.0	5	0	49.0	1.021	44.5–52.8	1.013	0.994	0.817
1	1.0	20	0	42.2	1.000	39.2–47.0	0.997	0.917	0.753
1	0.5	10	0	48.8	1.021	44.5–52.0	1.014	0.985	0.806
1	2.0	10	0	43.8	1.001	39.8–48.2	1.000	0.934	0.766
0.5	1.0	10	0	46.0	1.003	39.5–51.5	1.003	0.976	0.880
2	1.0	10	0	46.0	1.012	42.2–49.2	1.011	0.920	0.634
1	1.0	10	2	45.2	1.042	42.0–49.2	1.042	0.972	0.785
1	1.0	10	5	45.2	1.083	43.2–48.0	1.082	0.982	0.786

them: deposit inflows against the yield offered, and loan requests against the rate, once the pool has operated through some changes in both.

2. *No data on default.* The 3, 5 and 10% books are scenarios. The loss model is a one-factor Gaussian model with independent years, constant exposure, loss given default of 100% and no seasoning (Vasicek, 2002; Claude Code, 2026b).
3. *Higher rates may raise default.* If an increase of one point in the loan rate raises the annual default probability by γ points, lenders’ long-run yield in the regime $EL(a) > ar$ bends back when $\gamma > (1 - f)/\kappa = 0.95$, where $\kappa = 1.05$ is the slope of expected loss in the default probability, and credit is rationed (Stiglitz and Weiss, 1981). While the reserve covers losses ($ar > EL(a)$) lenders’ yield $ua(1 - f - r)$ is unaffected, but the lock fails once $\kappa\gamma > r$: $\gamma > 0.43$ at $r = 45\%$ and $\gamma > 0.62$ at 65%. Neither value has been measured.
4. *Static, steady-state analysis.* The model compares equilibria. It does not model the path of deposits and loans, entry of competing pools, or the response of other protocols’ rates.
5. *Safety-first with one criterion, and two participation rules.* The probability of a negative return in a year of the first ten is one of several reasonable criteria. It ignores the size of the loss and treats a loss of one cent as one of the whole deposit. The two rules of Section 3 bracket the horizon question but do not settle it: a discounted utility over an uncertain holding period, with entry cohorts that find the reserve at different levels, would be a third.
6. *Liquidity assumes staggered maturities and on-time repayment.* Synchronised maturities

Table 14: Three books. Columns: annual default probability (%); premium (bps); loan rate (%); expected loss (%); share that covers expected loss r_c (%); floor at no margin and $r \leq r_c$ (%); share at which lenders earn only the funding rate at the production rate (%); volume-maximising share, base case (%); shares within 1% of the maximum (%); volume at 45% and at 65%, relative to the volume at r_c .

PD	prem.	APR	EL	r_c	floor	r for e	$\hat{r}$	within 1%	45%	65%
3	600	10.33	3.06	29.6	8.16	50.7	33.5	28.7–38.0	0.940	0.649
5	800	12.33	5.15	41.8	10.24	58.7	46.0	41.2–50.2	1.006	0.784
10	1400	18.33	10.55	57.6	15.65	72.2	64.0	59.5–67.5	0.947	1.016

Table 15: Lenders' return and borrowers' earned credit at the production rate. Columns: reserve share r (%); lenders' long-run return (2) (%); interest net of the reserve share, $ua(1 - f - r)$ (%); return above the funding rate (%); growth rate of a borrower's limit, ra (%); years to double (Proposition 6); dollars of interest per dollar of earned credit, $1/r$.

r	return	net interest	above e	growth	doubling	$1/r$
30.0	6.10	7.34	1.77	3.70	18.7	3.33
41.8	6.10	6.10	1.77	5.15	13.5	2.39
45.0	5.76	5.76	1.43	5.55	12.5	2.22
50.0	5.24	5.24	0.91	6.17	11.2	2.00
55.0	4.72	4.72	0.39	6.78	10.2	1.82
60.0	4.19	4.19	-0.14	7.40	9.4	1.67
65.0	3.67	3.67	-0.66	8.01	8.6	1.54

change the shape of (10) but not its bound for a full run. Late loans open a gap in collections that no rate correction captures, and defaults change the claim rather than its timing (Section 7).

7. *The reserve's cash is idle.* The model leaves the cash that the reserve holds outside the lending base, as the contract does. Lending it would raise interest and risk, and the analysis here does not cover it.
8. *Synthetic accounts.* Nothing here is evidence about the welfare of borrowers.

11. Implications for the choice of the share

The analysis changes how the question is posed, and in this scenario supports six statements. The objective throughout is loan volume, which is not borrowers' welfare.

1. *The best share is a plateau, not a point, and its position depends on how lenders value the horizon.* Under the long-run rule the plateau for the 5% book extends from 41.2% to 50.2%, with its peak a few points above r_c , and it moves with the book across the three books of Table 14. Under a common ten-year holding period it extends from 40.2% to 50.2%, and over five years from 29.0% to 46.2%: the reserve's build-up is a cost to the lenders who fund it, which weighs more the shorter their horizon. The share that covers expected loss is itself a function of the book: 41.8% at 5%, 29.6% at 3%, 57.6% at 10%.
2. *Beyond the plateau the cost rises quickly.* Each point above r_c costs lenders 0.10 points of long-run yield per year; the equilibrium rate must rise, and volume falls by about 1.3% per point near 50% in the yield-only scenario (formula (8)) and faster above. Above 58.7% lenders earn less than the funding rate at the production rate.
3. *Seed capital can be cheaper protection than share, over the same horizon.* It protects the first years, when lenders bear most of the risk and reserve funded by interest has not yet built up.

Over ten years it costs less than the equivalent share at the funding rate and about the same at a 15% hurdle for the protection of 45%, and less at both rates for the protection of higher shares (Table 9); the result is conditional on the horizon, the hurdle and a seed that is never recovered.

4. *The premium and the share move together, and volume still falls.* If the share rises and the premium does not, volume falls to what lenders will fund (Table 12, “fixed”). The incidence formula gives the rate at which the market clears again; at that rate volume is lower than before, by (8), not restored.
5. *The share is a weak lever for growing borrowers’ credit.* Dues add a few percent a year to a limit (Table 15).
6. *With on-time repayment the pool is liquid at the scale of one loan term.* A stricter promise to lenders bounds utilisation and raises the rate floor; late loans add a delay of their own.

These are results of the model and its scenarios. The choice of a share in production should also weigh what the model leaves out: the protocol’s governance, the risk of the issuer’s lines, the cost and availability of seed capital, and evidence from the pool itself.

12. Conclusion

A reserve share sets a wedge between what borrowers pay and what lenders receive, and a reserve that is never released makes the wedge permanent once it exceeds expected loss. We have shown how the wedge divides between the two sides of the market, why a share below expected loss is free, why a share a few points above it is worth having when lenders care about losing principal, and why a larger share reduces lending. The pool’s liquidity follows from the term of its loans. What remains is measurement: the pool’s own flows can pin down the elasticities on which the position of the plateau depends.

Disclosure of AI authorship. This paper was written by Claude Code, an AI agent made by Anthropic, working with the project’s human operator, who set the research direction and asked for a demand and equilibrium model of the reserve share. The simulation of lenders’ returns reuses the loss model of the calibration code written by AI agents working with the project; the equilibrium, safety and liquidity computations are this paper’s own. All of it is public and reproducible.

Data and code availability. The loss model and its calibration, at commit b725a85, are in `analysis/credit_risk` of <https://github.com/scottonchain/microcredit-contract>. The script that produces every table, curve and number of this paper, with the commands to reproduce them, is in `lending-equilibrium/scripts` of <https://github.com/scottonchain/microcredit-theory>.

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Appendix A. Proofs

Proof of Lemma 1. If $R_{t-1} + uar \geq uL_t$, then $R_t = R_{t-1} + uar - uL_t$ and $\ell_t = 0$. Otherwise $R_t = 0$ and $\ell_t = uL_t - R_{t-1} - uar$, so again $R_t - R_{t-1} = uar - uL_t + \ell_t$. Hence $Y_t = ua(1 - f - r) - \ell_t = ua(1 - f) - uL_t - (R_t - R_{t-1})$, and summing gives the identity.

Let $\xi_t = uar - uL_t$, $S_t = \sum_{s \leq t} \xi_s$ and $\mu = u(ar - EL)$. Unrolling the recursion, $R_T = \max\{k + S_T, \max_{1 \leq j \leq T}(S_T - S_j)\}$. The ξ_t are independent and bounded, so $\sum_t \text{Var}(\xi_t)/t^2 < \infty$ and Kolmogorov’s strong law gives $S_n/n \rightarrow \mu$ almost surely (Billingsley, 1995). If $\mu > 0$, $\min_j S_j$ is finite, $R_T = S_T + O(1)$ and $R_T/T \rightarrow \mu$. If $\mu \leq 0$, then for any $\epsilon > 0$ almost surely $|S_n - n\mu| \leq \epsilon n$ for large n , so $S_T - S_j \leq (T - j)\mu + \epsilon(T + j) \leq 2\epsilon T$ and $0 \leq R_T \leq \max\{k + S_T, 2\epsilon T\}$, whence $R_T/T \rightarrow 0$. In both cases $R_T/T \rightarrow \max\{\mu, 0\}$. The loss rates also obey the strong law, $T^{-1} \sum L_t \rightarrow EL$. Dividing the identity by T , $T^{-1} \sum Y_t \rightarrow ua(1 - f) - uEL - \max\{uar - uEL, 0\} = y(a, r)$. $\square$

Proof of Proposition 1. $y(a, r) = \min\{u[a(1-f) - \text{EL}], ua(1-f-r)\}$ is the minimum of two strictly increasing continuous functions of a (as $r < 1-f$), so it is strictly increasing, and its inverse at $e+x$ is the maximum of the two inverses, which is (6). Identity (5) is $y = e+x$ rearranged. The first branch of (6) does not depend on r and is the larger one exactly when $a_0(x)r \leq \text{EL}$, that is, for $r \leq \bar{r}(x)$. The second is strictly increasing and convex in r , with derivative $(e+x)/(u(1-f-r)^2) = \underline{a}/(1-f-r)$, and unbounded as $r \rightarrow 1-f$. $\square$

Proof of Proposition 2. Let $\Phi(a) = y(a, r) - e - \psi(V^d(a))$, with $\Phi = -\infty$ where $V^d(a) \geq \bar{V}$. The function $y(\cdot, r)$ is continuous and strictly increasing. Since V^d is decreasing and ψ increasing, $\psi(V^d(a))$ is nonincreasing, so Φ is strictly increasing. At $a = 0$, $y = -u\text{EL} < e$, so $\Phi < 0$. As $a \rightarrow \infty$, $y \rightarrow \infty$ while $\psi(V^d(a))$ is bounded above by its value at any finite a , so $\Phi > 0$ for large a . By continuity Φ has a root, unique because Φ is strictly increasing; at it, $y = e + \psi(V^*)$ with $V^* = V^d(a^*)$, which is (7) after rearranging as in Proposition 1. For $r' > r$, $y(a, r') \leq y(a, r)$, so $\Phi(a^*(r); r') \leq 0$ and $a^*(r') \geq a^*(r)$; the inequality is strict where $a^*(r)r > \text{EL}$ because y then strictly decreases in r . Then $V^* = V^d(a^*)$ and $\psi(V^*)$ are nonincreasing. $\square$

Proof of Proposition 3. In the regime $ar > \text{EL}$, $y = ua(1-f-r)$, so $d \ln y = d \ln a - dr/(1-f-r)$. Along the equilibrium, $d \ln V^s = \varepsilon d \ln y$ and $d \ln V^d = -\eta d \ln a$ are equal. Hence $(\varepsilon + \eta) d \ln a = \varepsilon dr/(1-f-r)$, which gives the first formula. Then $d \ln V^* = -\eta d \ln a$ and $d \ln y^* = d \ln a - dr/(1-f-r) = -\eta/(\eta + \varepsilon) \cdot dr/(1-f-r)$. $\square$

Proof of Lemma 2. If $\ell_t > 0$ then $R_t = 0$ and $Y_t = ua(1-f) - uL_t + R_{t-1}$; if $\ell_t = 0$ then $Y_t = ua(1-f-r) > 0$. So $Y_t < 0$ if and only if $uL_t > ua(1-f) + R_{t-1}$. By induction on (1), on every path each R_t is nondecreasing in r , k and a . The right side of the inequality is nondecreasing in all three, so the event shrinks as any of them rises, and so does its probability. $\square$

Proof of Proposition 4. (1) Equilibrium solves $\Phi(a; r) = \ln V^s(y(a, r), p_H(r, k, a)) - \ln V^d(a) = 0$. Existence of an equilibrium and its uniqueness follow as in Proposition 2: Φ is strictly increasing in a (by Lemma 2, p_H falls as a rises, so supply rises), equals $-\infty$ for $y \leq e$ and is positive for large a . The root $a^*(r)$ is continuous in r , so V^* is continuous; and since $(1-f-r)a^* \geq e/u$ by (5), $a^* \rightarrow \infty$ and $V^* \rightarrow 0$ as $r \rightarrow 1-f$, so V^* attains its maximum. Take a maximiser r_0 . If $a^*(r_0)r_0 \geq \text{EL}$ we are done. Otherwise let $r_1 = \text{EL}/a^*(r_0) > r_0$ and $a = a^*(r_0)$. Then $\max\{\text{EL}, ar_1\} = \text{EL} = \max\{\text{EL}, ar_0\}$, so the yield is the same, and $p_H(r_1, k, a) \leq p_H(r_0, k, a)$ by Lemma 2, so supply at a is no smaller. As supply increases in a (through y and p_H) and demand decreases, the equilibrium rate at r_1 is no larger, so $V^*(r_1) \geq V^*(r_0)$. Since r_0 is a global maximiser, $V^*(r_1) = V^*(r_0)$, and because V^d is strictly decreasing this forces $a^*(r_1) = a^*(r_0) = a$. Hence r_1 is also a maximiser and $a^*(r_1)r_1 = ar_1 = \text{EL}$, which is what was claimed. (The inequality $a^*(r_1) \leq a$ alone would give $a^*(r_1)r_1 \leq \text{EL}$, the wrong direction; the equality of volumes is the step that closes it.) (2) At an interior maximiser $dV^*/dr = 0$, hence $da^*/dr = 0$, so $\partial\Phi/\partial r = 0$: $\partial \ln V^s/\partial r = -s_y ua + s_p \pi_r = 0$, using $\partial y/\partial r = -ua$ in the regime $ar > \text{EL}$, which is (9). (3) By Lemma 2, p_H is nonincreasing in k , so supply is nondecreasing in k at every (a, r) , and the equilibrium rate is nonincreasing and volume nondecreasing. $\square$

Proof of Proposition 5. The pool's cash is $(1-u)D + R$ and lenders' claims total D . A run of WD takes the cash first, leaving $(W - q_0)^+D$ to be paid from repayments. Loans of total uD mature uniformly over T days, so repayments arrive at uD/T a day, all to the queue. The time to pay $(W - q_0)^+D$ is $T(W - q_0)^+/u$. For $W = 1$ this is $T(u - c)/u = T(1 - c/u)$. Setting $w(W) \leq \bar{w}$ and solving for u gives (11). $\square$

Proof of Proposition 6. Interest paid on a drawn limit $\ell + d$ is $a(\ell + d)$ a year, of which the share r is added to d . So $\dot{d} = r a (\ell + d)$ and $\ell + d = \ell e^{r a t}$. The limit doubles when $r a t = \ln 2$, and one dollar of dues takes $1/r$ dollars of interest. $\square$